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Improving national fertility evaluations by accounting for the rapid rise of embryo transfer in US dairy cattle

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ABSTRACT

Dairy producers have improved fertility of their herds by selecting bulls with higher conception rate evaluations. This research was motivated by the rapid increase in embryo transfer (ET) use to 11% of recent births and >1 million total births, with >5 times as many ET calves born in the United States in 2021 compared with just 5 yr earlier. Historical data used in genetic evaluations are stored in the National Cooperator Database. Recent records in the national pedigree database revealed that only 1% of ET calves have corresponding ET records in the breeding event database, 2% are incorrectly reported as artificial inseminations, and 97% have no associated breeding event. Embryo donation events are also rarely reported. Herd years reporting >10% of calves born by ET but less than half of the expected number of ET breeding events were removed to avoid potential biases. Heifer, cow, and sire conception rate evaluations were recalculated with this new data set according to the methods used for the official national evaluations. The edits removed about 1% of fertility records in the most recent 4 yr. Subsequent analysis showed that censoring herd years with inconsistent ET reporting had little effect on most bulls except for the highest ranking, younger bulls popular for ET use, and with largest effects on genomic selection. Improved ET reporting will be critical for providing accurate fertility evaluations, especially as the popularity of these advanced reproductive technologies continues to rise.

Key words: embryo transfer, dairy cattle, fertility evaluations, conception rate

INTRODUCTION

Reproductive problems are one of the top reasons for culling in US dairy herds (Norman et al., 2021). Cow fertility data have been available from the DHIA for many years but were not routinely evaluated until 2003. At that time, evaluations were administered by the USDA Animal Improvement Program Laboratory (**AIPL**) researchers who developed format 5 as an avenue to collect reproductive records from US herds (CDCB, 2006). These include information like inseminations for cows, heat detection, breedings, pregnancy checks, synchronized breeding events, embryo donation, embryo implantation, sire and donor/recipient dams of embryo, and were stored in the AIPL database. Format 5 was developed in 2002 for efficient internet exchange and database storage of up to 20 reproductive events per lactation for millions of cows and heifers each month. For embryo recipient events, the sire and donor dam are stored in 2 separate segments to reduce the maximum format length from 1,047 to 737 bytes, reduce transfer time, and reduce storage required. The AIPL took over the calculation of estimated relative conception rate (Clay and McDaniel, 2001) in May 2006, which characterized fertility by the nonreturn rate, an indirect measure of fertility estimated by the proportion of cows that are not re-bred during 70 d after first insemination. Around this time, a massive research effort was launched to improve the accuracy of male fertility evaluations and broaden the scope to also include female fertility (Kuhn et al., 2004; VanRaden et al., 2004; Kuhn et al., 2006; Kuhn and Hutchison 2008; Kuhn et al., 2008). In August 2008, sire conception rate (**SCR**) was implemented, and heifer conception rate (**HCR**) and cow conception rate (**CCR**) were similarly developed and released in January 2009 (Norman et al., 2008). Since 2013, the newly reformulated Council on Dairy Cattle Breeding (**CDCB**) inherited responsibility for the data stewardship and administration of evaluations, whereas the Animal Genomics and Improvement Laboratory (formerly AIPL) remains as their research partner. Genomic evaluations for HCR and CCR are now provided to producers on a weekly basis by CDCB and SCR 3 times per year to address the high demand for accurate fertility evaluations to guide farm breeding programs.

Reproductive technologies have advanced significantly because these conception rate evaluations were

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developed and, consequently, dairy herd reproductive management is changing. Commercial embryo transfer (**ET**) began in the 1970s and today is becoming increasingly popular in herds desiring to increase their rate of genetic progress (Hasler, 2014; Moore and Hasler, 2017). Part of the attraction of ET can be explained by substantially reduced generation intervals among habitual ET users. Embryos purchased in high enough volumes could cost as little as $100 per embryo (Bowden and Thomas, 2022), putting ET at a competitive price level to AI services, which are approximately $15 per conventional service and $30 for sexed semen (Kaniyamattam et al., 2017). Benefits from ET may be more than twice those of AI due to higher conception rates and providing selection gains for breeding value (**BV**) instead of transmitting ability (Kaniyamattam et al., 2017).

With embryo transfer conception is not actually occurring, and so inconsistent reporting of ET calves and breeding events could bias fertility trait evaluations in the population. The United States has previously imposed edits on data that exclude known ET donors and reported recipients in our evaluations, and only 3 Interbull-participating countries account for ET by excluding ET records (Interbull Centre, 2021). This research was motivated by industry member requests to investigate anecdotal reports of bulls with large changes in EBV for conception rate and discussions on the increase in the ET popularity. In April 2022, new edits better accounting for the discrepancies in ET reporting were implemented (CDCB, 2022). In this paper, we detail the research supporting these changes, explore their effect on SCR, HCR, and CCR for 5 common dairy breeds in the context of bull age, bull genetic merit, bull popularity for ET, and overall bull popularity, and discuss the critical need to improve high-quality reproductive management reporting.

MATERIALS AND METHODS

Data Types and Availability

No human or animal subjects were used, so this analysis did not require approval by an Institutional Animal Care and Use Committee or Institutional Review Board. The data used in national evaluations comprise phenotypes, pedigrees, and genotypes from dairy herds across the United States, and they flow into the National Cooperator Database maintained by CDCB from a variety of sources including Dairy Records Processing Centers (**DRPC**), breed associations, the Interbull Centre (Uppsala, Sweden), the National Association of Animal Breeders (Madison, WI), and genomic laboratories and nominators (Miles and Parker Gaddis, 2022). For this study, data were extracted from the National Cooperator Database in December 2021, representing the most current information available on ET use in the United States. Embryo transfer status was determined from the codes stored in pedigree records that report the donor dam, birth records for calving ease and stillbirth evaluations that report the recipient dam, and reproductive records stored in herd management software, and collected and transmitted by DRPC. The pedigree and birth records indicate whether the animal in question was born as a singleton, multiple (e.g., twins), artificially split embryo, clone from nuclear transfer, or from ET and report the ID of their donor and recipient dam. Most herds doing ET also genotype the resulting heifer calves, so the pedigree records are verified and the correct genetic (donor) dam is known. These birth records were restricted to female calves only given that males are rarely recorded. The reproductive records of the recipient dam include breeding event codes for natural service, conventional AI, AI with sexed semen, and embryo donation or implantation. The data used in this study include 90,298,072 pedigree and 37,433,064 birth records dating from 1950. Reproductive records have only been available since 2003 and include 152,326,650 breeding events. Embryo transfer has only become popular in recent years and over 1.2 million ET calves have been reported, comprising 1.6% of total birth records. Embryo donation or implantation account for 345,028 of the breeding events, comprising only 0.2% of total reproductive records available.

Embryo Transfer Reporting Error Rates

An initial concern was that ET calves were being incorrectly coded as resulting from AI, accounting for the large discrepancy in ET calves born and ET breeding events reported. This could arise from a scenario where a cow was inseminated, came back into heat, and then received an embryo, but the only record to reach us is of the AI event and not the embryo implantation. A subsequent pregnancy exam could mistakenly confirm the AI to be a success instead of the unreported ET. To test this theory, reported breeding event types were matched with recorded birth types for a small subset of recent data (records from August 2016 through November 2020). Format 5 reproductive records report the ID of the female becoming pregnant, and optionally the sire and donor dam ID in the case of ET, whereas birth records report recipient dam ID and pedigree records report donor dam ID only. Recipient dam ID were used to link the birth type with the reproductive records of the recipient dam in question. Donor ID from the reproductive records were cross-checked with the donor ID listed in the pedigrees associated with the embryo

to ensure accurate matching. Potential misreporting of breeding events was identified by calculating the gestation length from date of breeding event to date of calving. Any records reflecting a gestation that was not within ±14 d from the breed average length were removed, with appropriate 7-d adjustments for the typical age of embryo at implantation. The number of ET calves corresponding to each breeding event type (AI, AI from sexed semen, natural service, embryo donation, and embryo implantation) were calculated and divided by the total number of calves born to determine the percentage of ET calves associated with each breeding event type.

Conception Rate Evaluations

Sire conception rate is computed within each breed Ayrshire, Brown Swiss, Guernsey, Jersey, Holstein, and Milking Shorthorn as follows:

$$y = HYS_D + P_D + SN_D + MY_D + A_D + SC_D + I_{SSR} + EI + AG_{SSR} + AIY_{SSR} + R_{SSR} + BV_D + PE_D + e,$$

where y is 1 for a confirmed success (validated by pregnancy check or resulting calving date) or 0 for a confirmed failure (if another breeding event is subsequently reported, subsequent calving date not within 14 d of expected, sold for reproductive problems, or designated "do not breed") for each insemination of a bull, as predicted by the fixed effects of herd-year-season for the dam (HYS_D), the parity of the dam (P_D), the service number of the dam (SN_D, defined as 1 for dam's first service of a parity, 2 for the dam's second service of a parity, and so on), standardized milk yield of the dam (MY_D), dam age (A_D), short cycle indicated by breeding ≤17 d after the time the dam was serviced (SC_D), service-sire inbreeding (I_{SSR}), expected inbreeding of the resulting embryo (EI), and the random effects of service-sire age group (AG_{SSR}), current year and AI company effect for the service-sire (AIY_{SSR}), residual service-sire effect (R_{SSR}), BV of the dam (BV_D), the permanent effect of the environment of the dam (PE_D), and residual deviance (e, for all models). The term "residual service sire effect" is used to avoid confusion with the published, full service-sire effect that includes the current age, expected inbreeding, and AI company effects. These models were developed from intensive research efforts to improve the accuracy of bull fertility evaluations and have been in use since 2008 (Kuhn et al., 2004, 2008; Kuhn and Hutchison, 2008; Norman et al., 2008). Only data from the most recent 4 yr are used and no heifer breedings nor crossbred cows are included. The BV_D is estimated only from the last 4 yr of data using pedigree relationships among cows to account for the merit of a sire's mates. This model does not include genetic effects but predicts the sire's total contribution to conception rate including his inbreeding, expected future inbreeding of embryos, age, residual effects, and current year AI organization effect.

The goal of SCR is to predict fertility of currently purchased semen by removing past environmental factors such as age, inbreeding, and company mean, and replacing those with the most recent estimates of the bull's environmental factors that can affect fertility. The published evaluations for SCR are expressed as a percentage and adjusted to the breed average, where a hypothetical bull with SCR of 2% is expected to have a conception rate 2 percentage points higher than the average conception rate for the AI bulls in that breed. Criteria for official publication of evaluations for bulls are ≥200 total breedings, ≥30 breedings during the most recent 12 mo for Ayrshire, Brown Swiss, and Guernsey; ≥300 total breedings, ≥100 breedings during the most recent 12 mo om ≥10 herds for Holstein; ≥200 total breedings, ≥100 breedings during most recent 12 mo in ≥10 herds for Jersey; and ≥100 total breedings, ≥10 breedings in most recent 12 mo in ≥5 herds for Milking Shorthorn. These differences reflect data availability for smaller breeds, which is also the reason this trait is computed separately for each breed rather than in an all-breed animal model. All data editing, quality assurance and control requirements, and methodologies related to official evaluations for SCR are publicly available through CDCB (CDCB, 2012).

Heifer conception rate and CCR are computed together in a multitrait model along with daughter pregnancy rate. These methods were developed from intensive research efforts on the use of field data and improving predictions of female fertility, and the multitrait model has been in use since 2014 (VanRaden et al., 2002, 2004, 2007, 2014; Kuhn et al., 2004, 2006; Wiggans and Goodling, 2005). Heifer conception rate describes a maiden heifer's ability to conceive and evaluations for HCR are calculated as follows:

$$y = HYS_H + A_H + PE_H + BV_H + e,$$

where y is defined as the percentage of inseminations that resulted in a pregnancy in a heifer and is standardized for the effects of region, breed, number of services, and breeding event type before inclusion in the model. Independent variables include the fixed effects of heifer's herd-year-season (HYS_H) and age (A_H) and the random effects of the heifer's permanent environment (PE_H) and the heifer's genetic effects on HCR (BV_H). Heifers have only 1 observation in y and would not need

a permanent environment effect except that in the multitrait model heifer and permanent environment effects are assumed correlated. Cow conception rate describes a lactating cow's ability to conceive, and evaluations are thus calculated as

$$y = HYS_C + P_C + A_C + PE_C + BV_C + e,$$

where y is the percentage of inseminations that resulted in pregnancies in each lactation for each cow and is standardized for the effects of region, breed, number of services, and breeding event type before inclusion in the model. The y variables for HCR and CCR are also weighted by number of attempts included. Independent variables include the fixed effects of cow's herd-year-season (HYS_C), parity (P_C), and age (A_C) and the random effects of the cow's permanent environment (PE_C) and the cow's genetic effects on CCR (BV_C). All available fertility records from 2003 onward are included and breeds are evaluated together in an all-breed animal model to allow crossbred cows to contribute to their sire's EBV. That is important for some smaller breeds where numbers of crossbred daughters can exceed purebred daughters. The SCR model might be converted to an all-breed model in the future for this same reason. Unknown parent groups are separate by breed and included in the EBV. All ancestors are retained in the pedigree, which allows tracing crossbred animals back to any combination of purebred ancestors. The *PE* effects for HCR and CCR are calculated separately but correlated by 0.10 in the multitrait model to account for a small environmental covariance between conception rate in heifers and cows (CDCB, 2020).

The EBV solutions for HCR and CCR are base-adjusted, divided by 2, and multiplied by 100 to obtain the published PTA expressed as percentages. An HCR of 2% implies that daughters of this bull are 2% more likely to become pregnant than the average in the base year 2015. A CCR of 2% implies that daughters of this bull are 2% more likely to become pregnant at each service than the breed average, also in base year 2015. Criteria for official publication of evaluations for bulls are at least 10 daughters with usable fertility data. A complete account of all other data editing and quality assurance and control requirements currently in use for the official evaluations are publicly available through CDCB (CDCB, 2020).

Edits for Embryo Transfer

The HCR, CCR, and SCR fertility evaluations continue to exclude all ET donor and recipient events if those events are reported. We developed new edits to account for unreported ET by censoring herd years with >10% of their calves born by ET but less than half of the expected ET breeding events given the number of ET calves born. This edit keeps most insemination records but removes those in herds most likely to introduce confounding bias due to unreported ET. Edits for the most recent partial year are based on the ET reporting of the previous full year. This new data set was used to compute SCR, CCR, and HCR with the same methodologies used for the official evaluations and described earlier. Pearson correlation coefficients ($\boldsymbol{\rho}$) between the new and old EBV were calculated, and the differences in each SCR ($\mathbf{SCR_{DIFF}}$), CCR ($\mathbf{CCR_{DIFF}}$), and HCR ($\mathbf{HCR_{DIFF}}$) were computed for each bull by subtracting the previous values from the new values generated with the proposed edit. The effects of this change were examined in the context of bull age, bull genetic merit, bull popularity for ET, and overall bull popularity. Correction of only the breeding events that result in pregnancy and a reported calf could create additional bias and inflate conception rate. Thus, successful AI or ET events are not generated later from the calving reports.

All comparisons were performed in SAS version 9.4 (SAS Institute Inc.) and data visualization was performed with R version 4.1.1 (R Foundation for Statistical Computing).

RESULTS AND DISCUSSION

As transfer technologies become more affordable, ET use has grown rapidly in the last several years with 11% of US dairy calves born in 2021 attributable to ET (Figure 1). However, the trend in ET breeding event reporting in format 5 records does not parallel the ET calf birth rate, and this lack of congruence can interfere both with national genetic evaluations and on-farm management of fertility. This discrepancy could be explained by several things, including embryo donation or implantation not being reported at all or ET being incorrectly reported as AI. Some of these errors can be eventually corrected when breed associations provide pedigree records, but those arrive 9 mo too late because fertility records are used in the evaluations when the breeding events occur, whereas pregnancy confirmations occur during gestation and birth types are recorded after calving.

The error rates in ET reporting in a subset of breeding event data from August 2016 to November 2020 are shown in Table 1. There were 35,100 reported AI breedings that corresponded to calves who were later reported born by ET, meaning that 0.32% of reported AI events were actually ET. Artificial insemination from

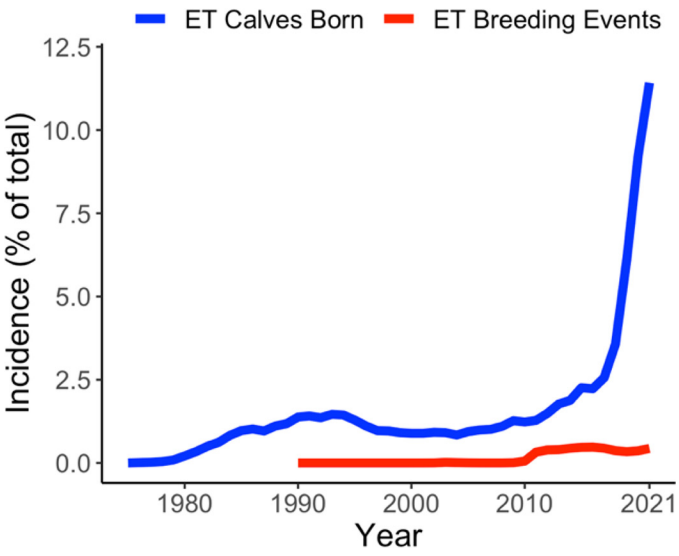


Figure 1. Embryo transfer (ET) usage in the United States. The percent of total calves born attributable to ET from 1975 through 2021 (blue) and the percent of reported breeding events related to ET from 1990 to 2021 (red) are shown.

sexed semen is coded separately, and we see that 1.25% of sexed semen usage reported actually corresponded to ET calves. This trend continues for natural service, where 0.35% of those matings corresponded to calves born by ET. However, even ET-related breeding events did not correspond to ET calves, with 0% of embryo donation events corresponding to ET births and only 1.25% of the 29,786 embryo implantation events corresponding to ET calves being born. Matching was often not possible because few embryo implantation records included the donor dam's ID and many calving ease records did not report the calf ID. Embryo implantation events could erroneously correspond to non-ET calves in a situation where ET was unsuccessful, and the recipient was subsequently inseminated by AI or natural service. Various studies report ET pregnancy rates ranging from 30% to 60% depending on many factors related to embryo quality as well as recipient selection and management (Ambrose et al., 1999; Hasler, 2001; Hansen, 2020), such that additional breeding events subsequent to implantations are not unlikely. We also observed that 535,290 of the ET calves born (10.1%) between August 2016 and November 2020 were not matched to any breeding events at all. These low rates suggest that ET donation and recipient events are not being obtained from most herds and, when obtained, are usually incorrect. The main remaining problem is standardizing the exchange of ET codes from on-farm software to DRPC. This will require the cooperation of software providers to modify their programs for the long-term storage and transfer of ET data.

The new edit resulted in the removal of 252 herd years for SCR, accounting for a 1.2% reduction in the total number of records included in the calculations. Numbers of lactation records used previously for HCR and CCR were 11,019,662 and 33,361,057 and those were similarly reduced by 237,414 (2.2%) and 323,618 (1.0%), respectively, by the new edit. Descriptive statistics on the changes in SCR, HCR, and CCR evaluations by breed are shown in Table 2. The high correlations and stable standard deviations of evaluations implied that variance components would not be affected and were not re-estimated in this research.

The median SCR_{DIFF} is close to zero for all breeds and the biggest changes were observed in Holstein bulls (range of −1.9 to +1.3 percentage points). We would expect a high correlation between SCR_{DIFF} and the frequency with which that bull is used for ET given that records were removed on the basis of ET usage, and the spread in SCR_{DIFF} by frequency used for ET is shown in Figure 2A. The SCR_{DIFF} converges to zero as more matings are reported (Figure 2B), which would suggest that the largest effects are in young bulls with fewer

Table 1. Embryo transfer (ET) reporting error rates[1]

	Birth type			
Breeding type	ET birth	Other[2]	Unmatched[3]	% ET
AI	35,100	11,060,000	6,360,421	0.32
AI (sexed semen)	14,833	1,175,795	727,909	1.25
Natural service	2,035	584,971	459,562	0.35
Embryo donation	0	13	4	0.00
Embryo implantation	372	29,416	18,373	1.25
Unmatched[4]	535,290	5,299,653	—	10.1

[1]Data from August 2016 to November 2020 were stratified by reported breeding type and matched to the reported birth types of the corresponding calves.
[2]Single birth, multiple birth, artificially split embryo, clone from nuclear transfer.
[3]Number of breeding events that were not matched to any birth at all.
[4]Number of births that were not matched to any breeding event at all.

Table 2. Summary statistics of the effect of new embryo transfer edits[1]

Item	Bulls compared	Change in evaluation (%)				Correlation[3]
		Minimum	Median	IQR[2]	Maximum	
SCR						
Ayrshire	6	−0.3	−0.1	0.4	0.3	0.98
Brown Swiss	44	−0.9	−0.1	0.3	0.6	0.99
Guernsey	14	−0.6	0.1	0.6	0.6	0.96
Holstein	1,379	−1.9	0.0	0.2	1.3	0.99
Jersey	132	−1.4	−0.1	0.3	0.9	0.99
Milking Shorthorn	1	0.0	0.0	0.0	0.0	—
HCR						
Ayrshire	22	−0.5	0.0	0.3	0.3	0.98
Brown Swiss	156	−0.6	−0.2	0.2	0.5	0.99
Guernsey	43	−0.8	−0.2	0.3	0.3	0.99
Holstein	13,807	−1.6	−0.1	0.3	3.7	0.99
Jersey	1,574	−1.6	−0.9	0.3	1.5	0.99
Milking Shorthorn	14	−0.4	−0.1	0.3	0.3	0.99
CCR						
Ayrshire	370	−0.8	−0.3	0.1	0.4	0.99
Brown Swiss	1,058	−1.5	−0.3	0.2	0.5	0.99
Guernsey	805	−0.8	−0.3	0.1	0.3	0.99
Holstein	47,262	−1.6	−0.3	0.2	1.4	0.99
Jersey	5,303	−1.8	−0.3	0.2	1.0	0.99
Milking Shorthorn	158	−0.9	−0.3	0.3	0.3	0.99

[1]The number of bulls compared, difference in evaluation, and correlation of old with new evaluation are shown for each sire conception rate (SCR), heifer conception rate (HCR), and cow conception rate (CCR).

[2]IQR = interquartile range.

[3]Pearson correlation coefficient between the new and old values.

daughter records. Embryo transfer usage is not strongly correlated with total number of matings among any breeds, except in evaluated Ayrshire bulls (Figure 2C). Ayrshire breeders may be following a more traditional program where ET represents a considerable investment and genomic predictions are not yet as accurate, and with fewer bulls available, proven bulls are prioritized for ET use. Other breeds may place more trust in genomics and prefer young bulls with high genetic merit for ET, explaining the negative correlations between total number of matings and percentage of ET usage. This trend may also reflect a forced policy enacted by AI companies of reserving use of elite new bulls for ET only to limit access to their sons. Indeed, we can confirm this theory by examining the changes in SCR by bull age and genetic merit (Figure 3). In Figure 3, we observe that younger bulls have greater SCR_{DIFF} and higher net merit. Nevertheless, the median SCR_{DIFF} for all breeds was very close to zero and the new SCR values were consistently highly correlated with the previous values ($\rho > 0.96$; Table 2).

The median HCR_{DIFF} was quite close to zero, with the largest changes seen in Holstein and Jersey (Table 2). Although Holstein bulls had the widest range in HCR_{DIFF}, their median value was still −0.1 percentage points, indicating that these ET edits are having overall small effect. The median HCR_{DIFF} for Jersey was −0.9, indicating that before the new edits the Jersey HCR on the all-breed scale was nearly 1 percentage point higher compared with Holsteins. A by-breed breakdown shows the differences in HCR for each bull by their popularity for ET (Figure 4A) and their overall popularity (Figure 4B). Bulls with the highest number of matings appear affected the least by the ET edit, again suggesting that these changes are negligible for widely proven bulls. Embryo transfer usage is not strongly correlated with total number of matings among any breeds (Figure 4C). This is likely because elite new bulls are being prioritized for ET use, and we can see quite clearly in Figure 3 that the biggest changes in HCR_{DIFF} are observed in young bulls with high genetic merit. Overall, these new HCR values were very highly correlated with the prior HCR estimates for every breed ($\rho > 0.98$).

The median CCR_{DIFF} was −0.3 percentage points for all 5 dairy breeds, suggesting that before this ET filtering edit all breeds were slightly overestimating CCR. Again, correlations between the prior CCR values and the new ones with the ET edit were consistently high ($\rho > 0.99$), suggesting that though some large changes were observed in individual bulls, the overall effect is low. A by-breed breakdown shows that CCR_{DIFF} tended to converge on zero as the total number of matings and the bull's popularity for ET increased (Figure 5A and B). For Ayrshire, Brown Swiss, and Guernsey, we see a moderate positive correlation between total number of matings and popularity for ET use, suggesting that

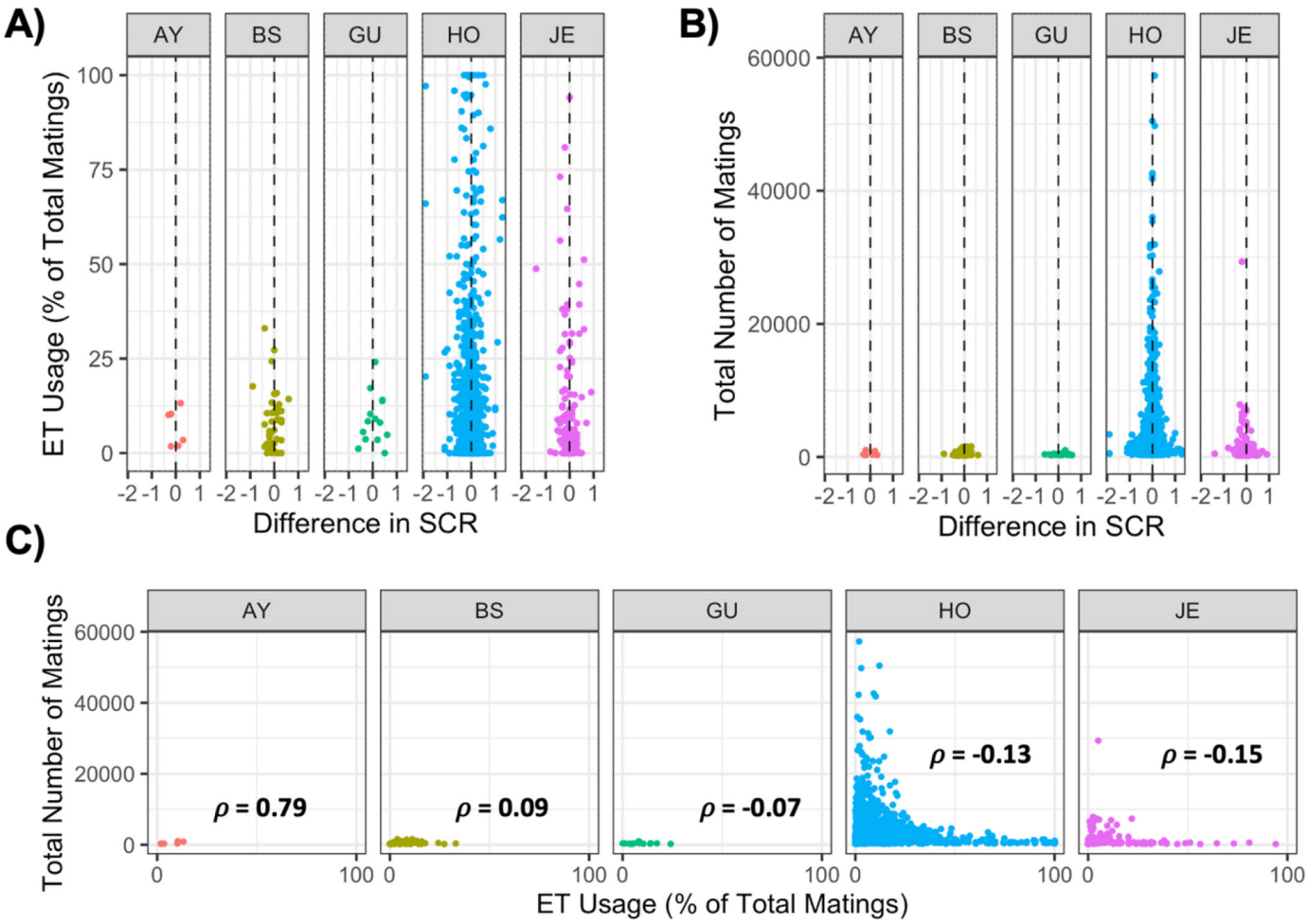


Figure 2. Differences in sire conception rate (SCR) by breed. Plotted against (A) percent of matings related to embryo transfer (ET) for each bull, (B) total number of matings for each bull, and (C) the correlation of total number of matings and popularity of each bull for ET. AY = Ayrshire; BS = Brown Swiss; GU = Guernsey; HO = Holstein; JE = Jersey; ρ = Pearson correlation coefficient.

those breeds are not limiting their ET use to elite new bulls as with HCR (Figure 5C). Holstein and Jersey have low correlations, which implies they are prioritizing elite new bulls for ET. Those breeds make up the bulk of the number of records, and we see in Figure 3 that overall, the biggest changes in CCR are in younger bulls with higher genetic merit. Cow conception rate and HCR are used in other total merit indexes including net merit, cheese merit, fluid merit, and grazing merit. They are also used in indexes provided by breed associations including the Holstein Association's Total Performance Index, the Jersey Performance Index, the Brown Swiss Association's Progressive Performance Ranking, as well as the Production Type Indexes for Red Dairy Cattle and Guernsey (CDCB, 2020). Inaccurate ET reporting could have a trickle-down effect, affecting important tools commonly used for overall dairy cow improvement.

This research was initiated after investigating anecdotal reports of young bulls whose conception rate estimates would change dramatically as more records were added, especially if their daughters were nearly all born by ET. The lack of reporting in some herds caused pregnancies to be falsely credited to a previous AI service instead of to the ET recipient event, which biased evaluations for the elite sires used in those herds. These changes would have major effects on AI companies and dairy producers alike, and so adapting fertility evaluations to account for changing reproductive management is critical to serving the dairy industry. The rapid increase in the ET calf birth rate (Figure 1) is not surprising when we consider a top genetic merit heifer whose embryos were collected starting at 7 mo old (Table 3). The maternal line continued this rapid turnover resulting in a remarkable 7 generations in 10 yr (average generation interval 16.7 mo). Some

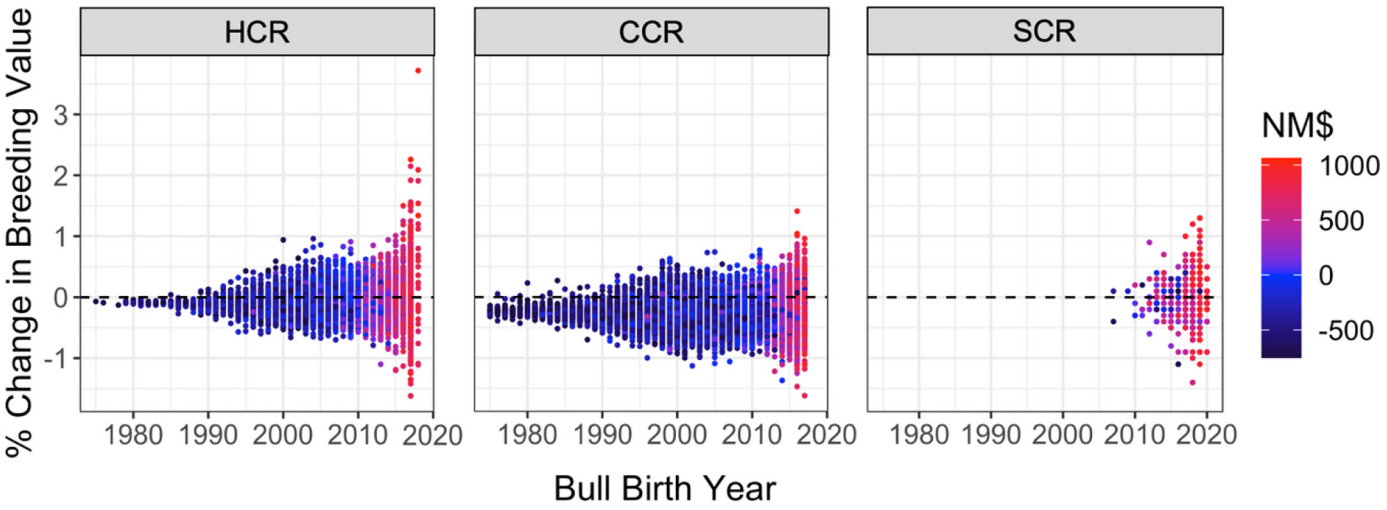


Figure 3. Effects of embryo transfer edits by bull age and genetic merit. The difference in each heifer conception rate (HCR), cow conception rate (CCR), and sire conception rate (SCR) are plotted by the bull's birth year and shaded by the net merit (NM$) of each bull.

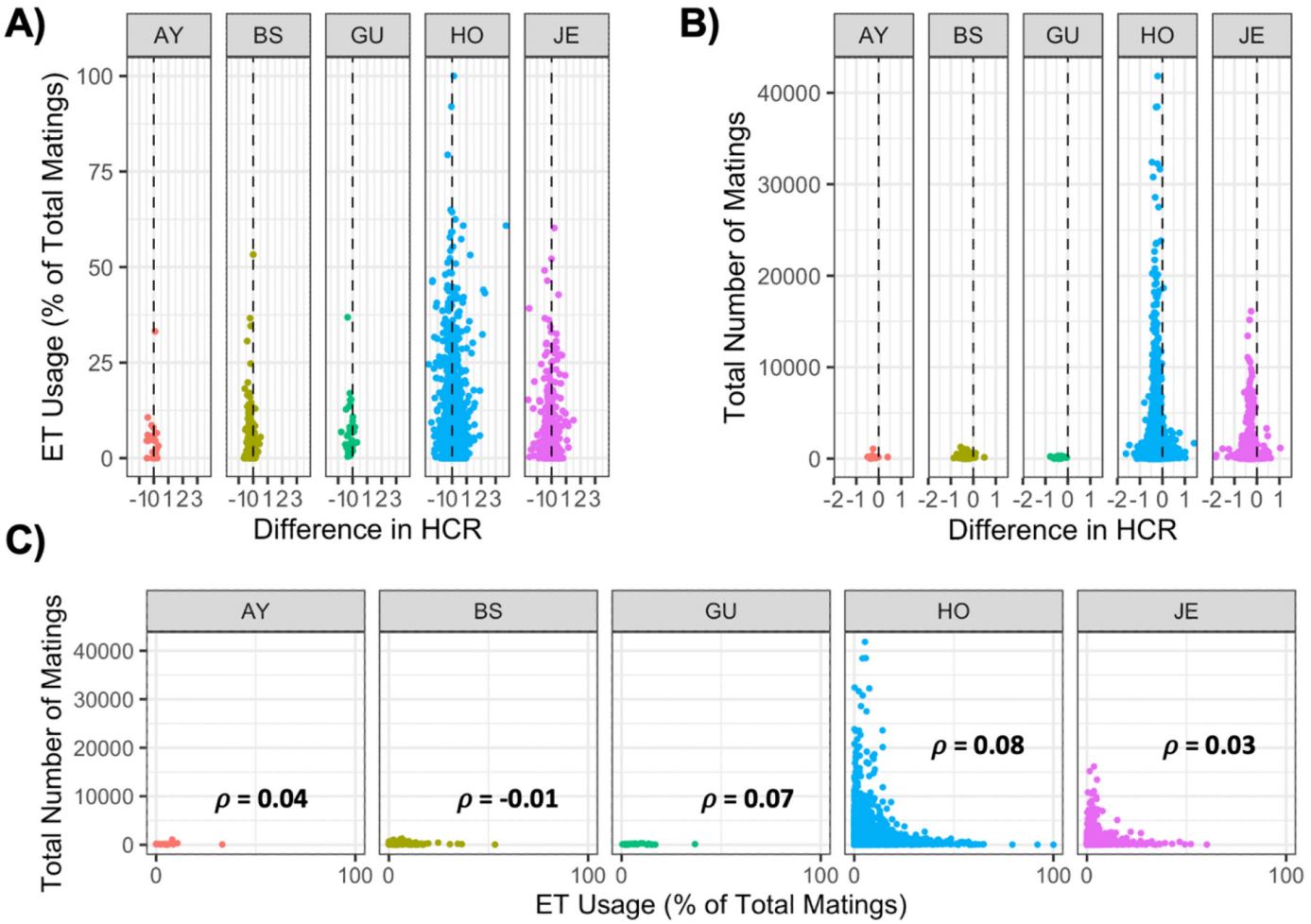


Figure 4. Differences in heifer conception rate (HCR) by breed. Plotted against (A) percent of matings related to embryo transfer (ET) for each bull, (B) total number of matings for each bull, and (C) the correlation of total number of matings and popularity of each bull for ET. AY = Ayrshire; BS = Brown Swiss; GU = Guernsey; HO = Holstein; JE = Jersey; ρ = Pearson correlation coefficient.

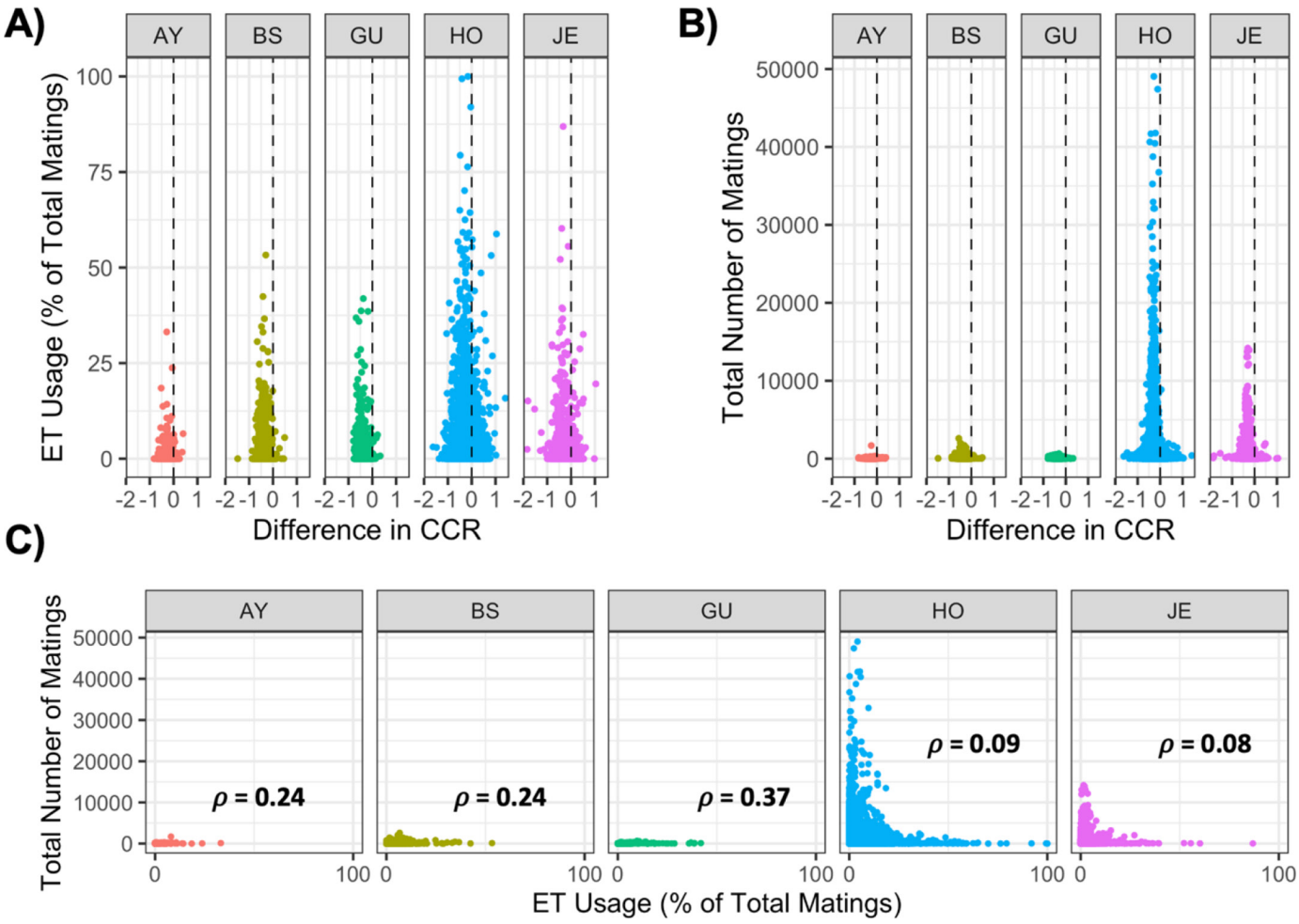


Figure 5. Differences in cow conception rate (CCR) by breed. Plotted against (A) percent of matings related to embryo transfer (ET) for each bull, (B) total number of matings for each bull, and (C) the correlation of total number of matings and popularity of each bull for ET. AY = Ayrshire; BS = Brown Swiss; GU = Guernsey; HO = Holstein; JE = Jersey; ρ = Pearson correlation coefficient.

generation intervals were likely the result of juvenile in vitro ET in which oocytes from pre-pubertal heifers were used (Hansen and Block, 2004; Kaniyamattam et al., 2017). Given the relative affordability and significant return on investment, ET is likely here to stay. Even before ET became popular for increasing the rate of genetic progress at the herd level, it dominated the bulls selected by AI companies (Figure 6). After rapid growth in the late 1970s, the percentages held steady at about 80% from 1990 to 2010 only to increase again to >90% after the genomics boom circa 2009.

Similar edits may be needed for other fertility traits like daughter pregnancy rate, gestation length, and early first calving. Beyond the scope of the edits described in this paper, progress in this area will be dependent on data quality and availability. It is likely that herd owners and managers who invest in an ET program also have good management of it, so the problem is that these herd records were not reaching the National Cooperator Database. Several years ago, the CDCB's Pursuing Data Quality team identified the primary obstacle to be on-farm recording and disseminated resources on correct ET entry into the most common herd management software. Further work to improve the standard transfer of high-quality ET records to DRPC is ongoing.

Acquiring high-quality ET records may help partition the genetic effects among male fertility and female contributions to conception rate and pregnancy rate. These would comprise details like multiple ovulation ET versus in vitro fertilization, fresh versus frozen embryos, embryo grade and stage, recipient synchrony, follicle stimulating hormone protocol, and number of degenerate embryos. The Beef Improvement Federation recently approved guidelines for using ET records and the necessary considerations in evaluation models (Beef

Table 3. Maternal line and genotyped embryo of dam HOCAN13913420

	Maternal generation							
Item	0	1	2	3	4	5	6	7
Birth year	2022	2020	2019	2018	2016	2015	2013	2012
Birth month	3	11	6	3	11	6	12	6
Age at embryo donation (mo)	—	7	8	6	7	8	9	9
Generation interval (mo)	16	17	15	16	17	18	18	—

Improvement Federation, 2021). These include alternative methodologies that account for the effects of both the donor (e.g., calf genetics) and recipient (e.g., classical maternal effects like birth weight) dams (Schaeffer and Kennedy, 1989; Suárez et al., 2015). These principles could also be adopted by the dairy industry with the establishment of new data pipelines. Until recently, unstandardized ET reporting had a minimal effect, but fertility evaluations will become less accurate if the ET calf birth rate continues to rise without a parallel increase in accurate ET breeding event reporting. As we move forward, it will be critical to evaluate the effectiveness of this edit, but this will be difficult to do without accurate ET reporting. One approach would be to evaluate the stability of the EBV for the bulls most affected by the edits with each release of the official tri-annual evaluations. Stable EBV would imply the edit is working, whereas highly fluctuating ones (which prompted the request for this research) would indicate that further edits are needed.

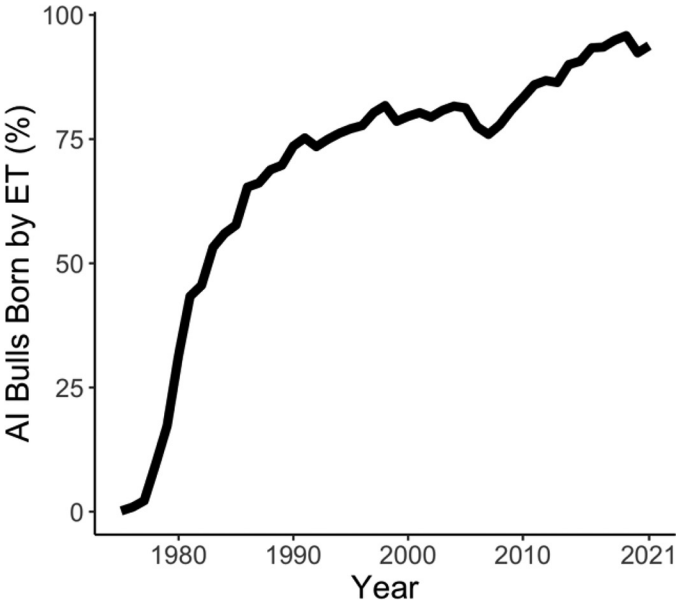


Figure 6. Embryo transfer (ET) usage for AI bulls in the United States. The percentage of bulls selected for AI that were born by ET from 1975 through 2021 is shown.

CONCLUSIONS

The rapid increase of ET is likely to continue as advanced reproductive technologies become more affordable. An investigation of SCR, HCR, and CCR shows that censoring herd years without consistent ET reporting has overall negligible effect, except for young bulls. Even though the correlations for each evaluation are high with small effects, the biggest changes are observed in elite new bulls, which have a huge influence on the breeding program. There is an urgent need to improve ET reporting to facilitate the delivery of accurate fertility evaluations.

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